

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The analysis of the tcpdump logs shows DNS traffic using the UDP protocol directed to port 53 on the DNS server. The client attempts to resolve the domain www.yummyrecipesforme.com, but the server responds with ICMP "Destination Unreachable – UDP port 53 unreachable".

This indicates that the DNS service is not reachable on port 53, preventing name resolution. As a result, the system cannot translate the domain into an IP address. The most likely issue is that the DNS service is not running or the UDP port 53 is being blocked by a firewall or misconfiguration.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident was first reported when users were unable to access the website and received a "destination port unreachable" error. To investigate, the security team used tcpdump to capture network traffic. The logs showed repeated UDP DNS queries followed by ICMP error responses indicating that port 53 was unreachable.

Key findings include:

- DNS queries are correctly sent via UDP
- No DNS response is returned
- ICMP errors confirm the service is unreachable on port 53

The most likely cause is a failure of the DNS service (service down, misconfiguration, or firewall blocking UDP/53). A DoS attack is possible but not confirmed by the available evidence.